

1.

For the flows represented by the following stream functions, determine the velocity components and check for the irrotationality,

(a) $\psi = xy$,

(b) $\psi = \ln(x^2 + y^2)$.

[Ans. (a) $u = x$, $v = -y$; irrotational flow

(b) $u = \frac{2y}{x^2 + y^2}$, $v = \frac{-2x}{x^2 + y^2}$; irrotational flow]

2.

In a two-dimensional incompressible flow over a solid plate, the velocity component perpendicular to the plate is $v = 2x^2 y^2 + 3y^3 x$, where x is the coordinate along the plate and y is perpendicular to the plate. Hence find out

(i) the velocity component along the plate

(ii) an expression for stream function and then verify whether the flow is irrotational or not.

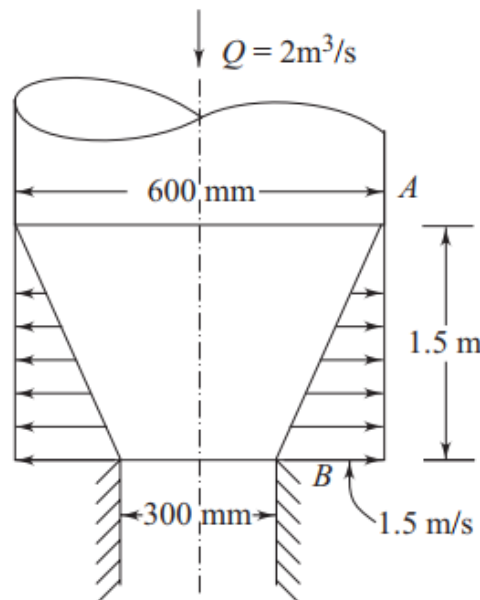
[Ans. $u = -\frac{4}{3}x^3 y - \frac{9}{2}x^2 y^2$, $\psi = -\frac{2}{3}x^3 y^2 - \frac{3}{2}x^2 y^3$; rotational]

3.



Water flows downward in a pipe of 600 mm diameter at the rate of $2 \text{ m}^3/\text{s}$. It then enters a conical duct (Fig. 4.37) with porous walls such that there is a radial outflow with flow velocity varying linearly from zero at A to 1.5 m/s at B . What is the rate of flow at B coming out from the conical duct.

(Ans. $0.587 \text{ m}^3/\text{s}$)



4.

Consider a cube with 2 m edges parallel to the coordinate axes located in the first quadrant with one corner at the origin. By making use of the velocity distribution $\vec{V} = 5x\vec{i} + 5y\vec{j} - 10z\vec{k}$, find out the flow through each face and show that no mass is being accumulated within the cube if the fluid is of constant density.

$$\begin{aligned} \text{[Ans. } x \text{ faces (faces } \perp \text{ to } ox\text{): } Q(\text{at } x = 0) &= 0, \\ Q(\text{at } x = 2) &= 40 \text{ m}^3/\text{s outflow} \\ y \text{ faces (faces } \perp \text{ to } oy\text{): } Q(\text{at } y = 0) &= 0 \\ Q(\text{at } y = 2) &= 40 \text{ m}^3/\text{s outflow} \\ z \text{ faces (faces } \perp \text{ to } oz\text{): } Q(\text{at } z = 0) &= 0 \\ Q(\text{at } z = 2) &= -80 \text{ m}^3/\text{s inflow}] \end{aligned}$$

5.

In a steady flow through a straight nozzle, the center line velocity changes from 1 m/s to 10 m/s in 0.3 m length. Determine the change in magnitude of convecting acceleration.

$$\text{(Ans. } 270 \text{ m/s}^2\text{)}$$

6.



A conical nozzle 0.5 m long converging from 0.4 m to 0.2 m diameter linearly is subjected to an outlet flow varying linearly from 0 to 1 m³/s in 10 seconds. Determine the total acceleration at the mid-length of the nozzle at the mid-time of variation. Assume uniform flow over each cross-section.

$$\text{(Ans. } 135.20 \text{ m/s}^2\text{)}$$

7.

For the following flows find the equation of the streamline passing through the indicated point:

- (i) $\mathbf{V} = 2x\mathbf{i} - y\mathbf{j} - z\mathbf{k}$ passing through (1, 1, 1)
- (ii) $\mathbf{V} = 4y\mathbf{i} - 3x\mathbf{j}$ passing through (-1, 2)
- (iii) $\mathbf{V} = -x\mathbf{i} + 2y\mathbf{j} + (3 - z)\mathbf{k}$ passing through (1, 1, 2)

$$\text{(Ans. } x^{1/2}y = 1 \text{ and } x^{1/2}z = 1, \text{ (ii) } 3x^2 + 4y^2 = 19, \text{ (iii) } xy^{1/2} = 1 \text{ and } x = 3 - z\text{)}$$

8.

The velocity in a fluid flow is expressed as

$$V = (xy + 2zt)i + (2y^2 + xyt)j + (12xy)k$$

Determine a_x , the x-component of the acceleration of the particle of fluid, at the point (1, 1, 2) at $t = 2.0$ seconds.

$$(Ans. a_x = 65 \text{ units})$$

9.

Verify whether the following flow fields are rotational. If so, determine the components of rotation about various axes.

$$u = xyz \quad (ii) \quad u = xy$$

$$v = zx \quad v = \frac{1}{2}(x^2 - y^2)$$

$$w = \frac{1}{2}yz^2 - xy$$

$$\left(Ans. (i) \text{ Rotational, } \omega_z = \frac{1}{2}z(1-x), \right.$$

$$\omega_x = \frac{1}{2}(z^2/2 - 2x), \omega_y = \frac{1}{2}y(x+1)$$

$$\left. (ii) \text{ Irrotational, } \omega_x = \omega_y = 0 \right)$$

10.

For the following set of velocity components verify whether the continuity equation is satisfied. If so, determine the (i) vorticity vector and (ii) acceleration vector at point A (1, 1, 1)

$$u = 2x^2 + 3y$$

$$v = -2xy + 3y^3 + 3zy$$

$$w = -\frac{3}{2}z^2 - 2xz - 9y^2z$$

$$(Ans. (i) \zeta_z = -(2y + 3);$$

$$\zeta_x = -(18yz + 3y), \zeta_y = 2z,$$

$$(ii) a_x = 32, a_y = -7.5 \text{ and } a_z = 93)$$

11.

Check whether the following sets of velocity components satisfy the continuity equation of steady incompressible flow.

$$(i) u = 4x + 2y - 3$$

$$v = 2x + 4y + 3$$

$$(ii) u = 4xy + y^2$$

$$v = 6xy + 3x + 2$$

$$(iii) u = 2x^2 + y^2$$

$$v = -4xy$$

$$(iv) u = x^3 + y^3$$

$$v = x - 3x^2y$$

$$(v) u = \frac{C(y^2 - x^2)}{(x^2 + y^2)^2}$$

$$v = \frac{-2Cxy}{(x^2 + y^2)^2}$$

(Ans. (i) not satisfied, (ii) not satisfied,

(iii) satisfied (iv) satisfied, (v) not satisfied)

12.

Calculate the unknown velocity component in the following so that the equation of continuity is satisfied.

$$\begin{array}{ll} \text{(i)} \quad u = \frac{2}{3}xy^3 - x^2y & \text{(ii)} \quad u = Aye^x \\ v = ? & v = ? \end{array}$$

$$\left(\begin{array}{l} \text{Ans. (i)} \quad v = -\frac{1}{6}y^4 + xy^2 + f(x), \\ \text{(ii)} \quad v = -\frac{Ay^2}{2}e^x + f(x) \end{array} \right)$$

13.

For the following sets of velocity components obtain the relevant stream functions.

$$\begin{array}{ll} \text{(i)} \quad u = 6y & \text{(ii)} \quad u = \frac{y^3}{3} + 2x - x^2y \\ v = 6x & v = xy^2 - 2y - \frac{x^2}{3} \end{array}$$

$$\begin{array}{l} \text{(iii)} \quad u = -A \ln x \\ v = A \frac{y}{x} + x^2 \end{array}$$

$$\left(\begin{array}{l} \text{Ans. (i)} \quad \psi = 3(y^2 - x^2) + C, \\ \text{(ii)} \quad \psi = \frac{y^4}{12} - \frac{x^2y^2}{2} + 2xy + \frac{x^3}{9}, \\ \text{(iii)} \quad \psi = -Ay \ln x - \frac{x^3}{3} + C \end{array} \right)$$

14

A steady, incompressible, two-dimensional velocity field is given by

$$u = \frac{y^3}{3} + 2x - x^2y$$

$$v = xy^2 - 2y - \frac{x^3}{3}$$

Does this represent an irrotational flow? If so, determine the relevant potential function which can represent this flow.

$$\left(\begin{array}{l} \text{Ans. flow is irrotational;} \\ \phi = \frac{xy^3}{3} + (2x^2 - y^2) - \frac{x^3}{3}y \end{array} \right)$$

15.

In a two-dimensional steady flow field, in a certain region of the x - y plane, the velocity component in the x -direction is given by $v_x = x^2$ and the density varies as $\rho = \frac{1}{x}$. Which of the following is a valid expression for the velocity component in the y -direction, v_y ? [2015]

- | | |
|------------------|-----------------|
| (a) $v_y = -x/y$ | (b) $v_y = x/y$ |
| (c) $v_y = -xy$ | (d) $v_y = xy$ |

Solutions: (c)

16.

A plane flow has velocity components $u = \frac{x}{T_1}$, $v = -\frac{y}{T_2}$ and $w = 0$ along x , y and z directions respectively, where $T_1(\neq 0)$ and $T_2(\neq 0)$ are constants having the dimension of time. The given flow is incompressible if **[2014]**

- (a) $T_1 = -T_2$ (b) $T_1 = -\frac{T_2}{2}$
(c) $T_1 = \frac{T_2}{2}$ (d) $T_1 = T_2$

Solution: (d)

17.

For a two dimensional flow field, the stream function ψ is given as $\psi = \frac{3}{2}(y^2 - x^2)$. The magnitude of discharge occurring between the stream lines passing through points (0,3) and (3,4) is **[2013]**

- (a) 6 (b) 3
(c) 1.5 (d) 2

Solution: (b)

18.

Potential function ϕ is given by $\phi = x^2 - y^2$. What will be the stream function (ψ) with the condition $\psi = 0$ at $x = y = 0$? **[2007]**

- (a) $2xy$ (b) $x^2 + y^2$
(c) $x^2 - y^2$ (d) $2 \times 2y^2$

Solution: (a)

19.

The velocity field for flow is given by $\vec{V} = (5x + 6y + 7z)\vec{i} + (6x + 5y + 9z)\vec{j} + (3x + 2y + \lambda z)\vec{k}$ and the density varies as $\rho = \rho_0 \exp(-2t)$. In order that the mass is conserved, the value of λ should be

[2006]

- | | |
|---------|---------|
| (a) -12 | (b) -10 |
| (c) -8 | (d) 10 |

Solution: (c)

20.

An inert tracer is injected continuously from a point in an unsteady flow field. The locus of locations of all the tracer particles at an instance of time represents

[2005]

- | | |
|-----------------|-----------------|
| (a) Stream line | (b) Path line |
| (c) Steam tube | (d) Streak line |

Solution: (d)

21.

A stream function is given by

$$\psi = 2x^2y + (x + 1)y^2$$

The flow rate across a line joining points $A(3,0)$ and $B(0,2)$ is,

[2005]

- | | |
|---------------|---------------|
| (a) 0.4 units | (b) 1.1 units |
| (c) 4 units | (d) 5 units |

Solution: (c)

22.

The circulation G around a circle of radius 2 units for the velocity field $u = 2x + 3y$ and $v = -2y$ is

[2005]

- (a) -6π units (b) -12π units
(c) -18π units (d) -24π units

Solution: (b)

23.

For a two-dimensional irrotational flow, the velocity potential is defined as $\phi = \ln(x^2 + y^2)$. Which of the following is a possible stream function, ψ , for this flow? [2003]

- (a) $\frac{1}{2} \tan^{-1}(y/x)$ (b) $\tan^{-1}(y/x)$
(c) $2 \tan^{-1}(y/x)$ (d) $2 \tan^{-1}(x/y)$

24.

The relation that must hold for the flow to be irrotational is [2000]

- (a) $\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} = 0$ (b) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$
(c) $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 v}{\partial y^2} = 0$ (d) $\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x}$

Solution: (a)

25.



The area of a 2 m long tapered duct decreases as $A = (0.5 - 0.2x)$ where 'x' is the distance in metres. At a given instant a discharge of $0.5 \text{ m}^3/\text{s}$ is flowing in the duct and is found to increase at a rate of $0.2 \text{ m}^3/\text{s}$. The local acceleration (in m/s^2) at $x = 0$ will be

- (A) 1.4 (B) 1.0
(C) 0.4 (D) 0.667

Ans. C

26.

In a steady flow through a nozzle, the flow velocity on the nozzle axis is given by $v = u_0(1 + 3x/L)i$, where x is the distance along the axis of the nozzle from its inlet plane and L is the length of the nozzle.

The time required for a fluid particle on the axis to travel from the inlet to the exit plane of the nozzle is

[GATE, 2007]

- (A) $\frac{L}{u_0}$ (B) $\frac{L}{3u_0} \ln 4$
(C) $\frac{L}{4u_0}$ (D) $\frac{L}{2.5u_0}$

Ans B

27.

For the continuity equation given by $\vec{\nabla} \cdot \vec{V} = 0$ to be valid, where $\vec{V}$ is the velocity vector, which one of the following is a necessary condition?

[GATE, 2008]

- (A) Steady flow (B) Irrotational flow
(C) In viscid flow (D) Incompressible flow

28.

Velocity vector of a flow field is given as $\vec{V} = 2xy\hat{i} - x^2z\hat{j}$. The vorticity vector at (1, 1, 1) is
[GATE, 2010]

- (A) $4\hat{i} - \hat{j}$ (B) $4\hat{i} - \hat{k}$
(C) $\hat{i} - 4\hat{j}$ (D) $\hat{i} - 4\hat{k}$

Ans. D

29.

Water is coming out from a tap and falls vertically downwards. At the tap opening, the stream diameter is 20 mm with uniform velocity of 2 m/s. Acceleration due to gravity is 9.81 m/s^2 . Assuming steady, inviscid flow, constant atmospheric pressure everywhere and neglecting curvature and surface tension effects,

the diameter in mm of stream 0.5 m below the tap is approximately [GATE, 2013]

- (A) 10 (B) 15
(C) 20 (D) 25

Ans. B

30

Consider a velocity field $\vec{V} = K(y\hat{i} + x\hat{k})$, where K is a constant. The vorticity Ω_z is [GATE, 2014]

- (A) $-K$ (B) K
(C) $-K/2$ (D) $K/2$

Ans. A

31.

A nozzle is so shaped that the average flow velocity changes linearly from 1.5 m/s at the beginning to 15 m/s at its end in a distance of 0.375 m. The magnitude of the convective acceleration (in m/s^2) at the end of the nozzle is _____. [GATE, 2015]

Ans. 540